

Annie Prasanna Manoharan

+1(479)301-1366 | annieprm0501@gmail.com | [Portfolio Website](#)

SUMMARY

I am a highly motivated person and problem-solver with a strong background in software development and test automation. Currently pursuing a Master's degree in Computer Engineering, which has provided me with a solid foundation in C++, SQL, data structures and algorithms. My hands-on experience in the industry at Soliton Technologies involved developing applications and test automation tools in C# and Python. This role, along with my research experience, demonstrates my strong analytical skills, ability to adapt quickly, and commitment to making impactful contributions.

SKILLS

Programming Languages: C, C#, C++, Python, Flask, SQL | **Databases:** MySQL, Oracle, MongoDB

Software Tools: Git, Bitbucket, Jira, LabVIEW, TestStand | **Methodologies:** Agile, SDLC

Soft Skills: Team Collaboration, Time Management, Adaptability, Ownership

EDUCATION

Master of Science in Computer Engineering
University of Arkansas, Fayetteville, AR | GPA 4.0/4.0

Aug 2024 - May 2026

INDUSTRY EXPERIENCE

Project Engineer, Soliton Technologies, India

June 2023 - June 2024

- Contract Engineer at **Dräger**: Developed a WPF desktop app in C# with integrated test automation (C#, Python, LabVIEW) to improve onboarding and boost ROI.
- Contract Engineer at **Medtronic**: Sole developer delivering MAVIS-based solutions; automated ~94% of manual test cases to meet demanding medical device compliance.

Intern at Soliton Technologies, India

July 2022 – May 2023

- Trained in LabVIEW and OOP with hands-on experience in automated testing using LabVIEW, TestStand, and MAVIS—Soliton's OOP-based test automation framework built on LabVIEW and Python for medical device V&V.

RESEARCH EXPERIENCE

Research Assistant at Artificial Intelligence and Computer Vision Lab, University of Arkansas

- Automated Chicken Gait Estimation** Sep 2024 – Feb 2025
Developed a real-time ML system using Meta's Detectron2 to detect leg diseases in chickens (in collaboration with Cobb). Used a dataset of 4,250 annotated images (2,646 train / 1,604 val) and the YOLO11n-Keypoint achieved AP50: 99.01% and mAP(50–95): 91.83%.
- Human Parsing Visualization for Virtual Try-On** May 2025 – Aug 2025
Built a tool to overlay and manually correct human parsing masks, improving training data quality. Leveraged GP-VTON (CVPR 2023) segmentation classes; processed 80k unlabeled and 12k labeled images.

TEACHING EXPERIENCE

- Programming Foundations (C++)** Aug 2025 – Dec 2025
Conduct lab sessions, office hours, and mentor 50+ students; create, review, and grade assignments focused on core programming fundamentals like control flow, loops, functions, arrays, and debugging.
- Database Management Systems** Jan 2025 – May 2025
Assisted in teaching a course with ~180 students by supporting course planning, content delivery, grading, and assessment development; provided guidance through office hours and discussions to clarify concepts.
- Programming (Assembly & VHDL)** Sep 2024 – Dec 2024
Instructed lab sessions for 35+ students across 3 sections, reinforcing theoretical concepts with hands-on implementation and offering technical support and debugging assistance.

PUBLICATIONS

(Journal) MiGa: Multi-Chicken Gait Assessment - MiGa, a multi-chicken gait assessment system that leverages computer vision and machine learning to automatically evaluate the gait of multiple birds simultaneously. (Under Review)

(Conference) DualFit: A Two-Stage Virtual Try-On via Warping and Synthesis - DualFit, a hybrid VTON pipeline that gives try-on results while maintaining high-frequency garment details, striking an effective balance between reconstruction accuracy and perceptual realism.(Accepted)

POSITIONS OF RESPONSIBILITY

- President of International Fellowship, a RSO that facilitates fun and educational opportunities for interaction between international students and residents of Northwest Arkansas. (2025-2026)
- Member of the Corporate Responsibility Team at Soliton Technologies in 2023-2024.